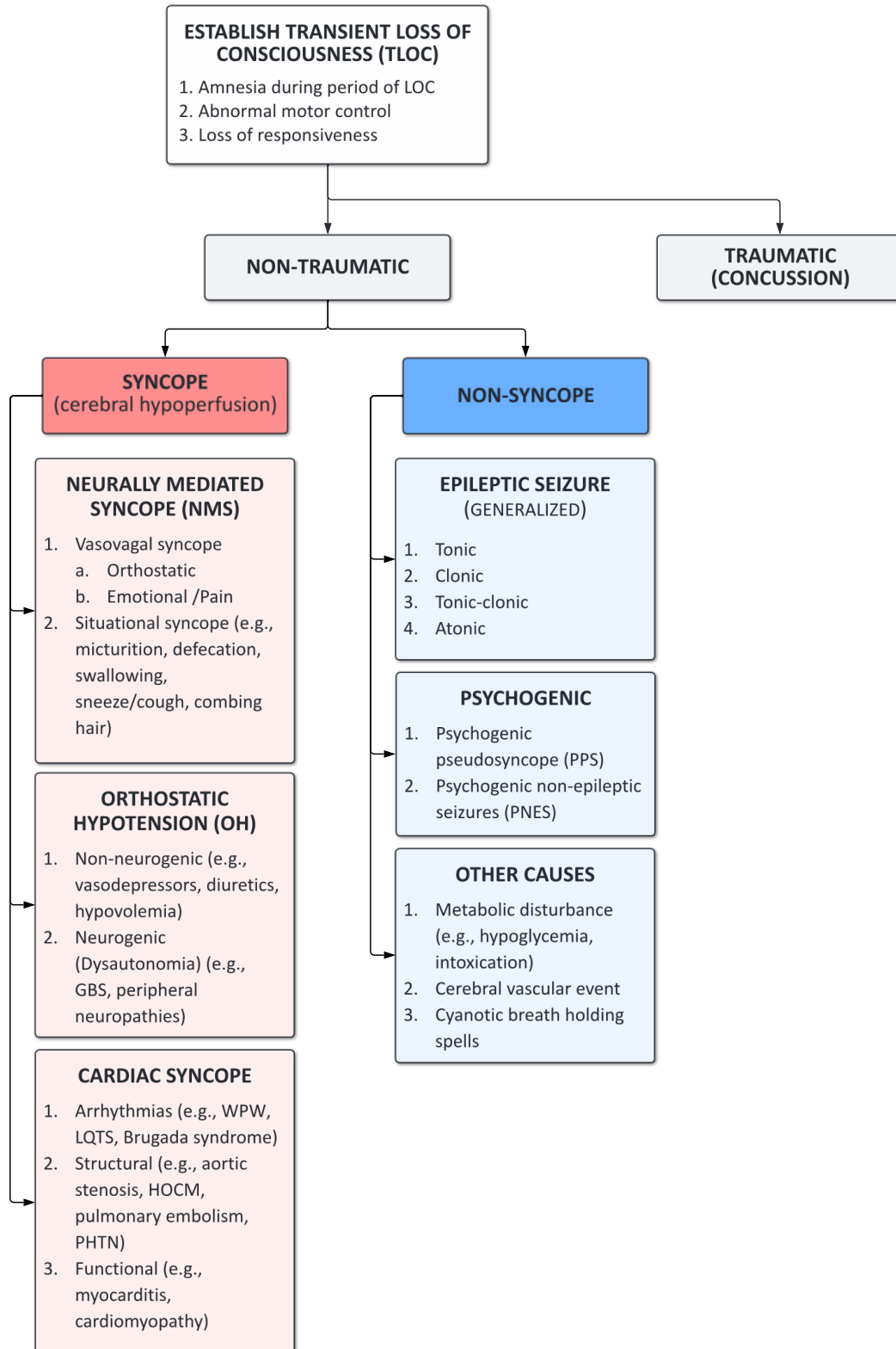
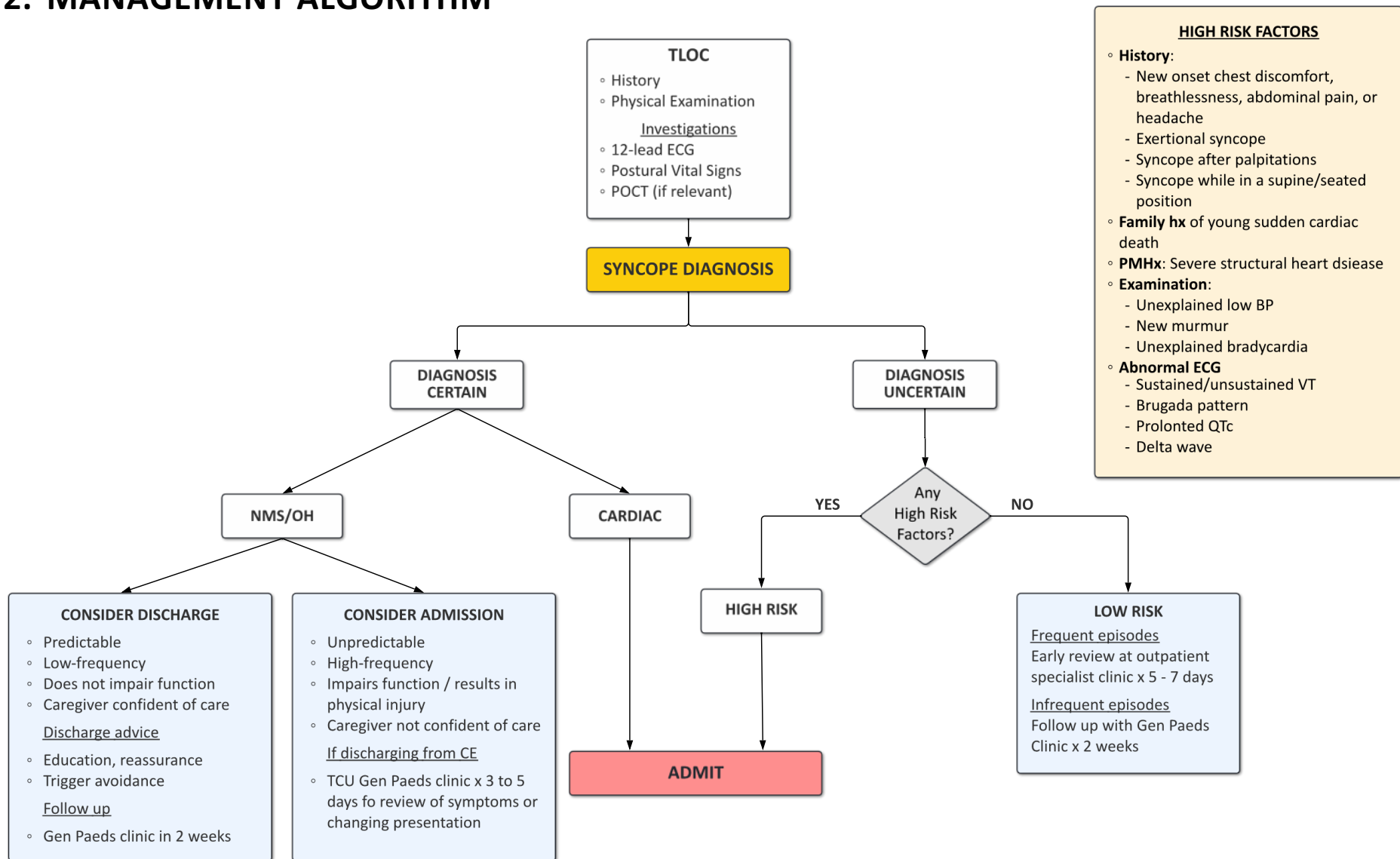


TRANSIENT LOSS OF CONSCIOUSNESS (TLOC) & SYNCOPE IN CHILDREN

1. APPROACH TO TLOC



2. MANAGEMENT ALGORITHM



3. DEFINITIONS

- Syncope ≠ Transient Loss of Consciousness (TLOC)
- **TLOC** is defined as a short state of real or apparent LOC with loss of awareness, characterized by amnesia for the period of unconsciousness, abnormal motor control, and loss of responsiveness
- **Syncope** is defined as TLOC due to cerebral hypoperfusion, characterized by a rapid onset, short duration, and spontaneous complete recovery
- **Presyncope** describes the state that resembles the prodrome of syncope without loss of consciousness

4. CLINICAL EVALUATION

- History should always be taken directly from first responders or first-hand witnesses

4.1 ESTABLISH LOC

- Amnesia during period of unconsciousness – any recollection of event by the patient?
- Were there purposeful movements observed by first responders?
- Loss of responsiveness

4.2 STRUCTURE OF HISTORY TAKING

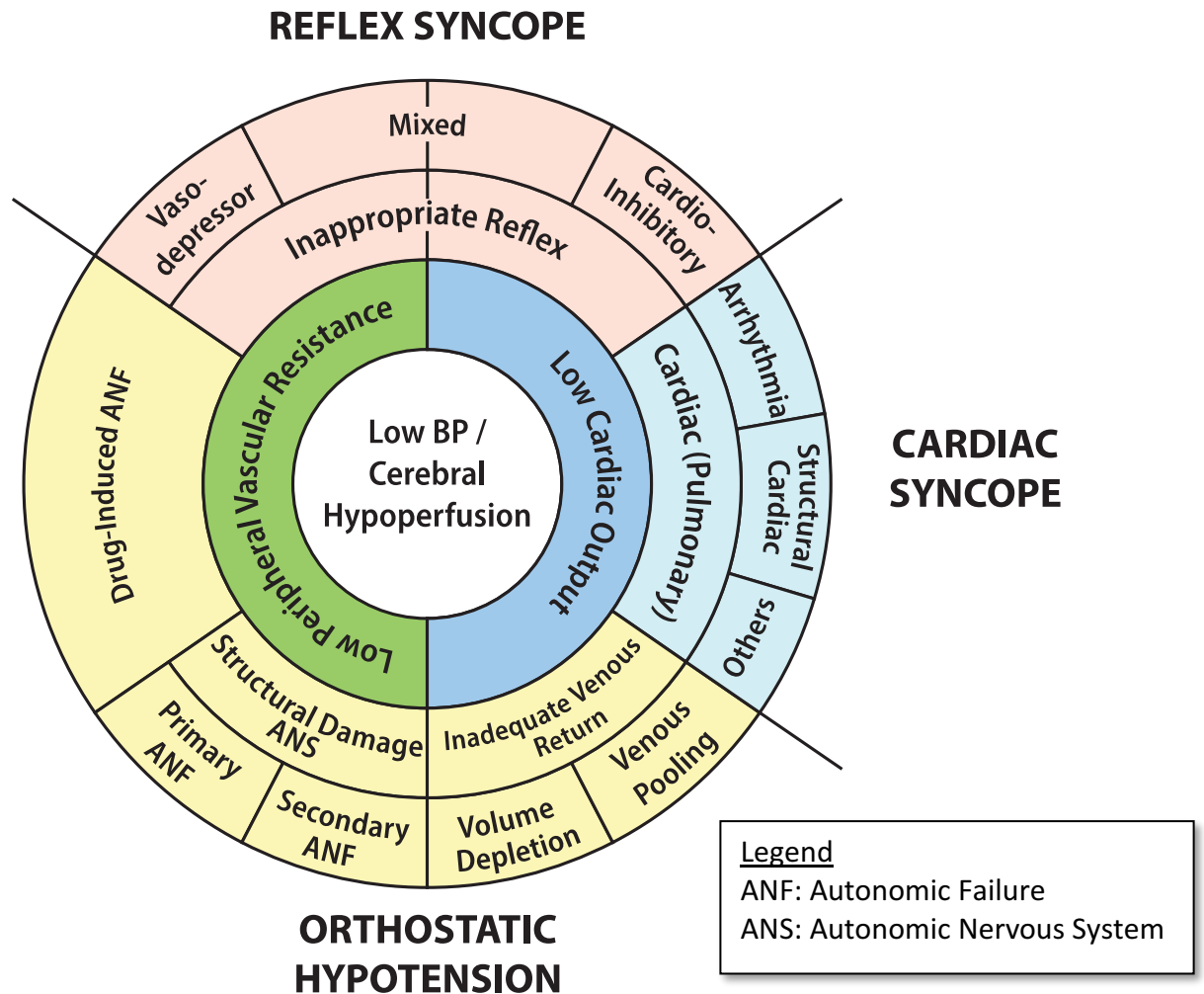
	EXAMPLES OF QUESTIONS TO ASK
PRE-EVENT	▪ Any triggers? ▫ Exertional vs. non-exertional ▫ Experienced pain or in an emotional state ▫ Related to posture ▪ Pre-syncope symptoms ▫ Chest pain, breathlessness, palpitations ▫ Pallor, diaphoresis ▫ Lightheadedness, flushed, warm, nausea? ▫ Darkening vision
DURING EVENT	▪ Duration of event ▪ Motor phenomenon (synchronized rhythmic jerking vs. pleomorphic muscle movement) ▪ Muscle tone (stiff vs. floppy) ▪ Skin manifestations (pale, cyanotic, cold, warm, diaphoretic) ▪ Soft features of seizure: tongue biting, urinary/bowel incontinence, physical injury
POST-EVENT	▪ Any post-event drowsiness/lethargy, confusion, headache

4.2.1 DIFFERENTIATING BETWEEN VARIOUS TYPES OF SYNCOPE AND NON-SYNCOPE

	SYNCPAL		NON-SYNCPAL (SEIZURE)
	CARDIAC	NMS/OH	
Posture	- While supine or in a seated position	<u>NMS</u> - With change in posture from sitting to standing <u>OH</u> - With prolonged standing	- Not related to posture
Triggers	- Exertional - Triggered by loud noise	- Unpleasant sight, smell, or pain - Prolonged standing - Cough, sneeze, defecation, micturition	- Intercurrent illness - Sleep deprivation - Non-compliance to medication
Prodrome / Pre-LOC	- Palpitations, chest pain, pallor, or cyanosis	<u>NMS (prolonged prodrome)</u> - Lightheadedness, diaphoresis, feeling warm, nausea - Persists as long trigger is present <u>OH</u> - No prodrome	- Brief aura
Features of convulsions (if present)		- Pallor - Tongue biting not common - Rarely has urinary incontinence - Pleomorphic muscle movement - Duration < 1 minute	- Cyanosis/normal - Tongue biting common - Lateral deviation - Urinary or bowel incontinence - Rhythmic convulsions - Duration several minutes
Symptoms with LOC	- Cold clammy skin, peripheral cyanosis	- Rarely has physical injury as patients prevent syncope by adopting a recumbent posture during prodrome	- Commonly associated with physical injury

	SYNCOPAL		NON-SYNCOPAL (SEIZURE)
	CARDIAC	NMS/OH	
Post-LOC		- Brief haziness, fatigue, diaphoresis, nausea - Quick return to baseline within a few minutes - Normal neurology	- Post-ictal confusion & drowsiness for more than 5 minutes - May have abnormal neurology
Risk factors	<u>Personal history</u> - History of exertional syncope or exertional dyspnea, chest discomfort - Extreme fatigue associated with exercise - History of structural heart problems or previous cardiac surgery <u>Family history</u> - Family history of young (< 50-years) sudden cardiac deaths (SCD) or sudden unexpected deaths (SIDS, car crash, drowning) - Family history of heart rhythm problems	- Started on new medications (e.g., diuretics) - Setup for dehydration	- History of epilepsy / neurodevelopmental problems - Recent titration in AEDs - Non-compliance to AEDs

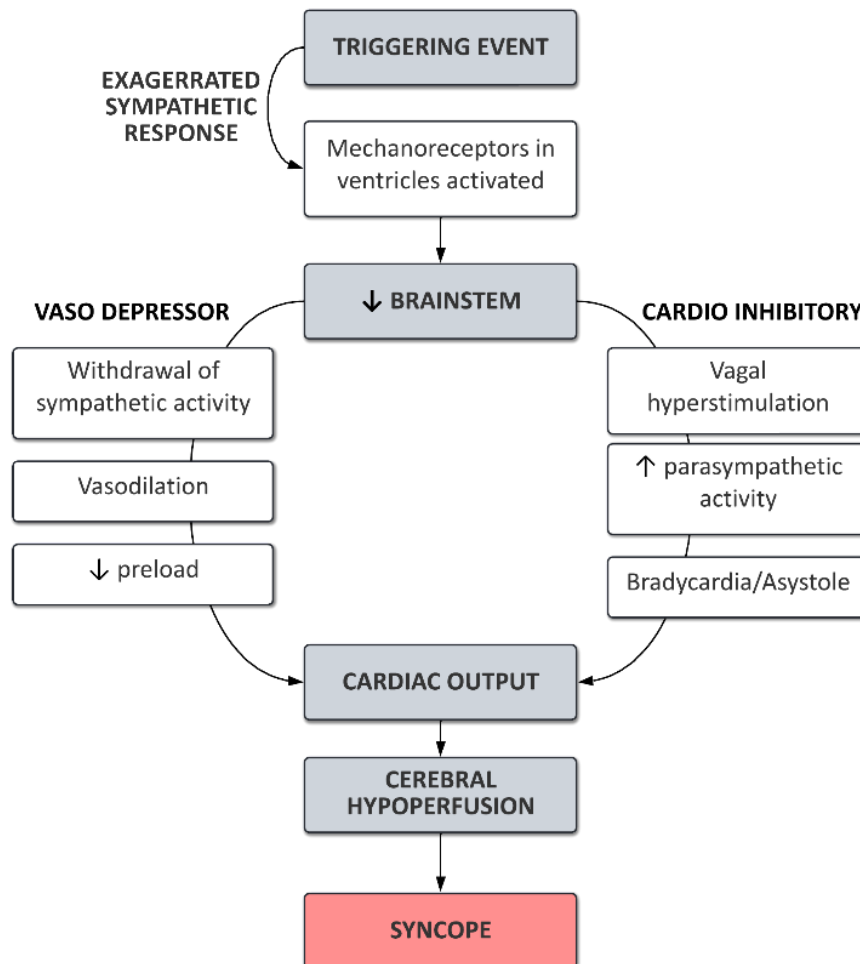
5. PATHOPHYSIOLOGY OF SYNCOPE



- There are 2 main mechanisms that lead to the final common pathway of syncope (low BP/cerebral hypoperfusion)
 - ↓ cardiac output
 - ↓ peripheral vascular resistance
- In syncope, both mechanisms often act together to a varying degree
- Both OH and reflex syncope span the 2 main pathophysiological mechanisms

6. TYPES OF SYNCOPE

6.1 NEURALLY MEDIATED SYNCOPE (NMS)



- Also called “reflex syncope” or “neurocardiogenic syncope”
- Mechanisms:
 - Vasodepressor ↓ sympathetic activity → ↓ pre-load → ↓ BP
 - Cardioinhibitory ↑ parasympathetic activity → Bradycardia / Asystole
 - Mixed Bradycardia + Hypotension

6.1.1 SUB-CLASSIFICATION

- Vasovagal syncope Pain, emotion, prolonged standing
- Situational syncope During a meal or post-prandial, micturition or defecation, cough, or sneeze, combing hair

6.1.1.1 PALLID BREATH HOLDING SPELLS

- Secondary to reflex bradycardia “cardioinhibitory” which results in cerebral hypoperfusion; not a result of cerebral hypoxia (as in cyanotic breath holding spells)
- Trigger is typically a minor trauma to the head/upper body that precedes LOC by 30 seconds (often obscuring the connection between the 2 events)
- Child stops breathing, becomes pale, diaphoretic & limp → this is followed by increased tone of trunk and extremities and often accompanied with incontinence and low-amplitude myoclonus
- Typically lasts less than one minute with confusion and sleepiness for several minutes after

6.2 ORTHOSTATIC HYPOTENSION (OH)

6.2.1 DEFINITION

- A sustained ↓ in BP within 3 minutes of standing, associated with symptoms
 - ↓ in SBP by >20 mmHg or DBP by >10 mmHg, OR
 - SBP < 90 mmHg (patients ≥ 10 years)
- Intermittent BP & HR readings while supine & during active standing for 3 minutes
 - 1 to 4 sets of readings recorded per minute, for 3 minutes
- Falls under the umbrella of Orthostatic Intolerance (OI) – inability to tolerate the upright posture because of signs and symptoms which are relieved by recumbence

6.2.2 NON-NEUROGENIC VS. NEUROGENIC (DYSAUTONOMIA)

NON-NEUROGENIC OH	NEUROGENIC OH (DYSAUTONOMIA)
- Common in children - No autonomic failure but incomplete ANS compensation for excessive non-autonomic stressors - Common causes: drugs (vasodepressors, diuretics) or hypovolemia	- Rare in children - Secondary to inadequate release of noradrenaline by the sympathetic nerves - May be seen during an acute infectious disease, or in peripheral neuropathies such as Guillain-Barré Syndrome (GBS)

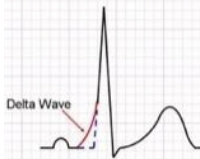
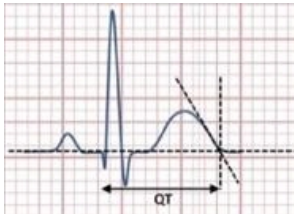
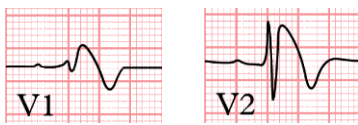
6.2.3 PRESENTATION

- Syncope upon standing
- Cerebral hypoperfusion can be exacerbated by concurrent disease or physiological states (e.g., post-exercise venous pooling, post-prandial, prolonged standing, after prolonged bed rest/deconditioning, anaphylactic shock)

6.3 CARDIAC SYNCOPE

- 3 categories of cardiac syncope:
 - Arrhythmias WPW, Long QT syndrome, Brugada syndrome
 - Obstruction Aortic stenosis, Hypertrophic cardiomyopathy, Pulmonary embolism, Pulmonary hypertension
 - Function Myocarditis

6.3.1 ECG CHANGES

	CLINICAL FEATURES	ECG CHANGES
Wolfe-Parkinson-White (WPW) syndrome	Reentrant circuit with narrow complex tachycardia → shorter diastolic filling time → ↓ stroke volume → ↓ cardiac output	 Shortened PR interval Widened QRS complex with a delta wave
Long QT syndrome (LQTS)	Torsades de pointes presents with <u>syncope</u>, <u>syncope followed by seizures</u>, <u>SCA</u>, or <u>SCD</u> Frequently misdiagnosed as epilepsy Triggers - Acute arousal (e.g., loud noise, emotion, physical exertion, or sudden waking up from sleep by alarm clock) - Medications - Electrolyte abnormalities Associated clinical features - Congenital deafness - Family member with definite LQTS - Unexplained SCD (< 30 years) in immediate family	 QT-interval = Start of QRS complex to where a tangent line drawn on the downslope of the T wave intersects with the baseline $QTc = \frac{QT}{\sqrt{RR \text{ (preceding)}}}$ QTc is prolonged if: - QTc > 0.44 (men) or - QTc > 0.46 (children & women)
Brugada syndrome	Unmasked by fever, drugs, hypokalemia, hypothermia, post-DC cardioversion. ECG + 1 clinical criteria - Documented VF or pmVT - Family hx of SCD @ < 45-years - Coved-type ECG in family members - Syncope	Coved ST segment elevation (>2mm) in > 1 of right precordial leads (V1 – 3) followed by a negative T wave. 

	CLINICAL FEATURES	ECG CHANGES
Myocarditis	Viral prodrome → non-specific symptoms (abdominal pain, vomiting, anorexia) Symptoms of heart failure (resting dyspnea & tachypnea, tachycardia, displaced apex, hepatomegaly) Investigations: - Myocardial injury: troponin I, CKMB - Markers of perfusion: Lactate, end-organ function (RP, LFT)	ST segment & T wave abnormalities, reduced voltage waveforms

6.3.2 MANAGEMENT OF CARDIAC SYNCOPE

Disposition

- Admit to HD/PICU: rhythm disorder, reduced cardiac function (myocarditis), critical obstruction (e.g., critical AS, severe CoA)
- GW with early review: obstructive lesion, non-specific syncope with high risk factors

Monitoring

- Patient placed in P2 for continuous cardiac monitoring while awaiting admission
- IV cannula

Investigations

- Blood gas with micro-electrolytes, Capillary blood glucose
- RP1, Cl, CO₂, Mg, iCa
- Troponin I, CK, CKMB, lactate
- FBC, CRP, ESR, LFT

Optimize patient's physiological state

- Maintain normothermia
- "Cardiac" electrolytes (K ≥ 4 mmol/L, Mg ≥ 1 mmol/L, iCa > 1.2 mmol/L)

Equipment for transfer

- Zoll defibrillator (+ 3-lead ECG), monitoring device
- Transfer kit (Emergency drugs, Intubation set)
- Oxygen tank
- Bag-and-Mask
- Sheet with precalculated doses of resuscitation drugs

6.4 NON-CARDIAC SYNCOPE (NMS/OH)

6.4.1 GENERAL ADVICE

- For patients with recurrent syncopal episodes
- Reassurance regarding benign nature
- Awareness & avoidance of predisposing states (e.g., dehydration, post-prandial, post-exercise)
- Avoiding triggers (e.g., hot crowded environments, prolonged standing)

6.4.2 PREVENTING SYNCOPE

- Early recognition of prodromal symptoms
- Supine position with legs elevated at onset of symptoms
- Counterpressure maneuvers
 - Cross legs & contract leg, abdominal & gluteal muscles
 - Arm tensing – gripping one hand with the other and simultaneously abducting both arms

6.4.3 LIFESTYLE MODIFICATIONS

- ↑ fluid intake
- Liberalizing salt intake

6.4.4 DISPOSITION

- Discharge if:
 - Predictable
 - Low frequency
 - Does not impair function
 - Caregiver confident of care
- Admit or early follow-up if:
 - Frequent & unpredictable syncopal episodes → at risk of self-injury
 - Caregiver not confident of care

6.5 SYNCOPE OF UNCERTAIN ETIOLOGY

- Patients with uncertain etiology should be stratified by:
 - High vs. low-risk features
 - Frequent vs. Infrequent events
 - Quality of life/function affected
- High Risk: Admit patient for observation
- Low Risk
 - Frequent episodes: TCU General Pediatric Clinic x 5 – 7 days
 - Infrequent episodes: TCU General Pediatric Clinic x 2 weeks

6.5.1 RISK STRATIFICATION

	HIGH RISK	LOW RISK
Syncopal Event	- New onset chest discomfort, breathlessness, abdominal pain, or headache - Syncope occurs during exertion, in a supine/seated position, after palpitations - Family history of young sudden cardiac deaths	- Prodrome of typical reflex syncope - Occurring after prolonged standing in a crowded place, after a meal - Triggered by cough, defecation, or micturition, unpleasant sight, smell, pain - Change in posture (supine to standing)
Past Medical History	- Severe structural heart disease (e.g., valvular lesions) - Coronary heart pathology (e.g., Kawasaki with coronary ectasia)	- Long standing history of recurrent syncope with low-risk features with similar features to current episode - No structural heart disease
Physical Examination	- Unexplained low blood pressure (for age) - New onset murmur - Unexplained bradycardia	- Normal examination
ECG	- Presenting rhythm: sustained or un-sustained VT - Brugada pattern (Type 1) - Prolonged QTC - Delta wave	- Normal ECG

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